Mechanism Description

Historical Exp-Data

Mechanism Analysis Agent

Parse Mechanism Rules

Analyze Historical Data

Synthesize & Infer

- Consistent — Increase Weight
- Conflict — Prioritize Data
- No Data — Infer from Mechanism

Weight Function Specification

```
{ "Temperature": { "center": 450, "range": [400, 500], "weight": 0.95 }, ... }
```

Construct Weight Function

Center Value → Gaussian Prior

-- Map to [0.3, 1.0]

$W(x)$

Weighted Bayesian Optimization

Train Surrogate Model

-- Gaussian Process

Bayesian Optimization Loop

1. Apply Knowledge Weight
2. Calculate Acquisition Function
   -- Adjusted Score = Acquisition(x) * W(x)
3. Select Next Point
   -- argmax(Adjusted Score)

Recommended Process Parameters